MD ARIF FOYSAL

Full-Stack Software Engineer | NestJS · Next.js · React · PostgreSQL | Machine Learning & AI Systems

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PROFESSIONAL SUMMARY

Full-stack Software Engineer with hands-on, production-level experience across the complete Software Development Life Cycle (SDLC) — requirement analysis, design, development, testing, deployment, and maintenance. Skilled in building and maintaining scalable web applications using NestJS, Next.js, React, and PostgreSQL, with additional backend experience in PHP and MySQL. Experienced in designing and consuming REST APIs, implementing role-based access control (RBAC), and optimizing relational database performance with TypeORM. Builds and evaluates machine learning and deep learning models (LSTM, GRU, Bi-LSTM) for time-series forecasting and predictive analytics, and automates workflows using n8n AI agents. Comfortable working in Agile/Kanban teams with Trello, using Git for version control, Docker for containerization, and CI/CD fundamentals for reliable software delivery, and managing live production deployments via FTP/SFTP (FileZilla). Published IEEE researcher in applied machine learning. Fast learner with a strong record of shipping reliable, well-architected software, seeking to grow into a professional full-stack / AI-integrated software engineering role.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, PHP, C++, C#, SQL

Frontend Development: React, Next.js, HTML5, CSS3, Tailwind CSS, Responsive UI Design

Backend Development: NestJS, Node.js, PHP, REST API Design & Documentation, Role-Based Access Control (RBAC), Server-Side Architecture

Databases & ORM: PostgreSQL, MySQL, TypeORM, Relational Database Design & Query Optimization

Machine Learning & AI: LSTM, GRU, Bi-LSTM, Time-Series Forecasting, Data Analysis & Preprocessing, SMOTE, SHAP/LIME (Explainable AI), Logistic Regression, Random Forest, XGBoost, n8n (AI Agent Automation)

DevOps & Tools: Git, GitHub, Docker, Postman, FileZilla (FTP/SFTP), CI/CD Fundamentals

SDLC & Process: Requirement Analysis, System Design, Development, Testing, Deployment, Maintenance, Agile/Scrum Sprints

Project Management: Trello (Kanban/Agile task tracking, sprint planning, backlog management)

PROFESSIONAL EXPERIENCE

Skyview Online Ltd **Feb 2026 – Present**
Software Developer *Dhaka, Bangladesh*

- Build a comprehensive HR Management System (HRMS) automating employee records, attendance tracking, and payroll processing end-to-end.
- Implement role-based access control (RBAC) to enforce secure multi-user management and data integrity across the platform.
- Optimize backend performance by refining REST API response logic and complex PostgreSQL queries within a NestJS architecture.
- Design and maintain a scalable backend architecture using PostgreSQL and TypeORM, focused on data consistency and query efficiency.
- Apply structured SDLC practices end-to-end — planning, development, testing, and release — tracking sprints and tasks in Trello (Agile).
- Manage production deployments via FileZilla (FTP/SFTP), handling file uploads, replacements, and live updates for software releases with minimal downtime.
- Developed and maintain the official company website of Skyview Online (skyviewonline.com).
- Built the ISP customer login portal (billing.skyviewonline.com) for customer billing and support.
- Developed two custom CRM platforms for GMAX BD, an IIG (International Internet Gateway) company, with a dedicated panel for each team: bw.gmaxbd.com and panel.gmaxbd.com.
- Designed the CRMs to serve the marketing, admin, transmission, billing, NOC, HR, support, legal, and call center teams across the organization.
- Platform supports 40+ team members and powers 24/7 customer support for 500+ customers.

PROJECTS

WhatsApp Notice Management System **2026**
Full-Stack Developer — NestJS, PostgreSQL

- Built a centralized full-stack system to manage and distribute notices across multiple WhatsApp groups via browser integration.
- Developed automated message broadcasting and real-time group synchronization features using event-driven backend logic.
- Designed and documented REST APIs and optimized relational database structures in PostgreSQL to support high message volume.

Attendance Intelligence System **2026**
Full-Stack Developer — PHP, MySQL

- Engineered a parsing tool to convert raw attendance logs into structured, actionable datasets.
- Generated automated reports for workforce tracking, salary timing calculations, and organizational efficiency analysis.

ML-Based Crypto & Stock Prediction **2025**
Machine Learning Project — Python, GRU, LSTM, Bi-LSTM

- Built time-series forecasting models to predict financial market trends using GRU, LSTM, and Bi-LSTM neural network architectures.
- Cleaned and analyzed financial datasets to improve prediction accuracy and identify historical market patterns.

RESEARCH EXPERIENCE

ML-Based Software Failure Risk Prediction **2026**
Co-Author (Research Project) — American International University-Bangladesh (AIUB)

- Proposed a machine learning framework using the NASA JM1 dataset for early software defect prediction.
- Implemented and compared performance metrics across Logistic Regression, Random Forest, and XGBoost classifiers.
- Utilized SMOTE for class imbalance handling and integrated SHAP/LIME for model interpretability (Explainable AI).
- Introduced a probability-based risk classification system to categorize software modules by failure risk.

PUBLICATIONS

A Comparative Study of GRU, LSTM, and Bi-LSTM for Financial Time Series Forecasting: Apple Stock and Bitcoin **2026**

IEEE Conference Publication (Co-Author)

- Conducted comparative analysis of GRU, LSTM, and Bi-LSTM deep learning architectures for financial time-series forecasting using Apple stock and Bitcoin datasets.
- Evaluated prediction accuracy and performance across deep learning models.
- DOI: 10.1109/QPAIN69676.2026.11545577

EDUCATION

American International University-Bangladesh (AIUB) **2022 – 2026**

BSc in Computer Science and Engineering — CGPA: 3.00

Ideal College, Dhanmondi **2019 – 2021**

HSC (Science) — GPA: 4.42